

2022 Ph H2 Q9

(a)

Calculate the energy of one photon of this radiation

Given:

• Wavelength $\lambda = 5.50 \times 10^{-7} \text{ m}$

• Use:

$$E = \frac{hc}{\lambda}$$

Where:

• $h = 6.63 \times 10^{-34} \text{ J s}$

• $c = 3.00 \times 10^8 \text{ m/s}$

$$\begin{aligned} E &= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{5.50 \times 10^{-7}} \\ &= \frac{1.989 \times 10^{-25}}{5.50 \times 10^{-7}} = 3.62 \times 10^{-19} \text{ J} \end{aligned}$$

(b)

Calculate the number of photons incident per second on the surface

Given:

• Power of radiation = $4.60 \times 10^{-3} \text{ W}$

• 1 photon carries $3.62 \times 10^{-19} \text{ J}$

Use:

$$\text{Number of photons per second} = \frac{\text{Power}}{E_{\text{photon}}} = \frac{4.60 \times 10^{-3}}{3.62 \times 10^{-19}} = 1.27 \times 10^{16}$$

(c)

State what is meant by wave-particle duality

✔ Wave-particle duality means that **electromagnetic radiation** (and indeed all quantum particles) can behave as both:

- **Waves**, showing interference and diffraction, and
- **Particles**, carrying energy in discrete packets called **photons**

Both wave and particle models are needed to fully explain quantum behaviour.

